

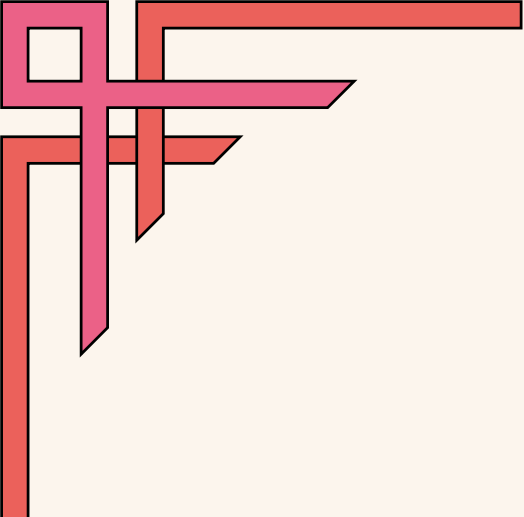
Week-2 Deliverable

**TICKET DATA
CLEANING, TAGGING &
SENTIMENT ANALYSIS REPORT**

Team-2

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CONTENTS

- 1. Introduction**
 - 2. Exploration & Understanding of Dataset**
 - 3. Data Extraction & Initial Assessment**
 - 4. Data Cleaning Process**
 - 5. Tagging & Categorization Strategy**
 - 6. Manual Sentiment Classification Approach & Evaluation**
 - 7. Sentiment Analysis Summary**
 - 8. Key Observations & Insights**
 - 9. Learning Outcomes**
 - 10. Conclusion**
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1. Introduction

The Week 2 “***Ticket Data Cleaning, Tagging & Sentiment Analysis Report***” activity focuses on transforming raw customer support ticket data into a clean, structured, and analysis-ready format. This phase emphasizes the critical role of data preparation in customer support analytics, where well-organized datasets enable accurate trend identification, performance evaluation, and evidence-based decision-making. During this week, Freshdesk ticket logs are extracted, reviewed, and systematically cleaned by standardizing priorities, categories, timestamps, and tags while removing irrelevant or sensitive information. In addition, a basic manual sentiment evaluation is applied to ticket subjects and descriptions to capture customer tone. Collectively, these steps ensure that the dataset is reliable, consistent, and fully prepared to support customer support insights and dashboard development in upcoming weeks.

Our aim is -

- To extract and understand the structure of raw Freshdesk customer support ticket data.
- To clean and organize ticket records by standardizing categories, priorities, statuses, and timestamps.
- To apply consistent tagging for improved issue categorization and clarity.
- To conduct a basic manual sentiment analysis to identify customer tone across tickets.
- To prepare a structured and reliable dataset that supports trend analysis and reporting in the subsequent week.



2. Exploration & Understanding of Dataset

This section provides an overview of the customer support dataset used in Week-2, describing its source, structure, scope, and key fields. The objective is to establish a clear understanding of the dataset before proceeding with data cleaning, tagging, and sentiment preparation activities.

2.1 Dataset Reference Overview

Dataset Type	Dataset Name (links)
Assigned Dataset	Freshdesk Customer Support Tickets Dataset
Filled Dataset (Week-1 Output)	Freshdesk Customer Support Tickets Dataset Filled

2.2 Dataset Description

The dataset used for this report is the “**Freshdesk Customer Support Tickets Dataset Filled**”, which represents anonymized customer support ticket records generated through a helpdesk system. Each record corresponds to an individual customer support request and captures essential operational details such as issue category, priority level, ticket status, assigned agent, timestamps, tags, and customer sentiment. The dataset reflects real-world support scenarios commonly encountered in customer service operations and serves as the foundational input for customer insights and reporting activities.

2.3 Connection with Week-1 Activity

The dataset used in Week-2 is a direct continuation of the *Week-1 “Ticket Mapping Report”* activity. In Week-1, the assigned “**Freshdesk Customer Support Tickets Dataset**” initially contained several analytical fields that were either empty or unstructured, including Category, Priority, Status, Tags, and Sentiment. As part of the ticket mapping exercise, each support ticket was carefully reviewed and these fields were manually populated using consistent classification rules based on the ticket subject, description, and resolution context.

This structured mapping process resulted in the “**Freshdesk Customer Support Tickets Dataset Filled**”, which serves as the primary input for Week-2. The completed dataset provides a standardized and well-defined foundation that enables further data cleaning, tagging refinement, and sentiment evaluation, ensuring readiness for accurate analysis and reporting in subsequent stages of the internship.

2.4 Dataset Size and Scope

- **Total number of tickets:** 30
- **Ticket range:** T011 to T040 (Unique Identifier)
- **Time coverage:** January 4, 2026 – January 10, 2026

2.5 Dataset Columns and Description

Column Name	Data Type	Description
Ticket ID	Text	Unique identifier assigned to each support ticket
Created Date	DateTime	Date and time when the ticket was created
Resolved Date	DateTime	Date and time when the ticket was resolved or closed
Agent	Text	Support agent assigned to handle the ticket
Customer	Text	Anonymized customer identifier
Subject / Short Description	Text	Brief description of the customer's issue or request
Category	Categorical	Type of issue reported (Technical, Billing, Feature Request, Account Management)
Priority	Categorical	Urgency level of the ticket (Low, Medium, High, Urgent)
Status	Categorical	Current or final state of the ticket (Open, Resolved, Closed)
Tags	Categorical	Short labels used to group similar issues
Sentiment	Categorical	Customer sentiment classified as Positive, Neutral, or Negative

2.6 Core Analytical Fields

The following fields are considered core analytical columns for Week-2 data preparation and subsequent analysis activities:

- **Ticket ID** – Serves as a unique identifier for each support ticket, enabling accurate tracking and record referencing.
- **Created Date & Resolved Date** – Capture the ticket lifecycle timeline, supporting time-based organization and resolution tracking.
- **Category (Issue Type)** – Defines the nature of the customer issue, allowing systematic grouping of similar support requests.
- **Priority** – Indicates the urgency level assigned to each ticket, supporting structured handling and response planning.
- **Status** – Represents the current or final state of the ticket within the support workflow.
- **Tags** – Provide additional contextual labels that enhance issue classification and improve filtering accuracy.
- **Sentiment** – Reflects the customer's expressed tone (Positive, Neutral, or Negative), supporting sentiment-based organization.

These core analytical fields play a critical role in supporting subsequent data cleaning, sentiment summarization, and reporting activities. By organizing tickets using standardized identifiers, timelines, issue categories, priorities, tags, and sentiment labels, the dataset becomes easier to interpret and manage. A clear understanding of these fields enables accurate tracking of customer issues, supports structured workflow analysis, and provides a reliable foundation for understanding customer support operations and service performance in later stages.

3. Data Extraction & Initial Assessment

This section outlines the process of exporting, inspecting, and evaluating the raw ticket data from Freshdesk. The goal is to understand the initial structure, identify potential inconsistencies, and prepare the dataset for further analysis.

3.1 Data Extraction from Freshdesk

Customer support ticket data is typically exported from Freshdesk using its **CSV export functionality**, which allows support teams to download ticket records for offline analysis. The exported CSV file contains all relevant ticket information, including creation and resolution timestamps, assigned agents, customer details, issue descriptions, and other analytical fields.

For this week, the dataset “**Freshdesk Customer Support Tickets Dataset Filled**” was used, which had already undergone initial mapping and categorization during Week 1. This dataset consists of **30 tickets (T011–T040)**, capturing real-world support interactions handled by agents over a defined period.

3.2 Raw Dataset Structure

We used python code in google colab (jupyter notebook) to inspect the raw dataset structure initially.

Upon initial inspection of the raw CSV, the following characteristics were observed:

- **Rows & Columns:** 30 × 11 (30 tickets and 11 fields).
- **Column Types:** All fields were imported as objects (`string` type) in Python, including date columns.
- **Potential Issues Identified:**
 - Dates were not in proper `datetime` format, requiring conversion.
 - Sentiment column contained inconsistencies such as extra spaces or duplicate entries (`'Neutral '` vs `'Neutral'`).
 - Category, Priority, Status, and Tags appeared consistent, but uniform formatting checks were needed.
 - No duplicate rows were present.
 - No missing values were found after initial inspection.

The raw structure reflected typical outputs from Freshdesk exports, which often require **data type conversions, standardization, and cleansing** to ensure accurate downstream analysis.

3.3 Initial Assessment & Python-Based Evaluation

Customer support ticket data exported from Freshdesk typically contains raw fields such as Ticket ID, Created & Resolved Dates, Agent, Customer, Subject/Short Description, and analytical fields like Category, Priority, Status, Tags, and Sentiment. These exports often contain common issues, which can affect downstream analysis:

- **Inconsistent tags** across tickets
- **Different naming conventions** for priority or status
- **Timestamp formatting variations**
- **Missing or duplicate values**

For the current dataset, a Python-based assessment was conducted to detect such potential issues. The following observations were made:

1. **Dataset Shape:**

- The dataset contains **30 tickets** and **11 columns** (30 × 11).

2. **Duplicate Rows:**

- No duplicate entries were present.

3. **Data Types:**

- All columns were initially of type **object**.
- Date fields (**Created Date** and **Resolved Date**) were stored as text rather than datetime objects, which could affect time-based calculations.

4. **Missing Values:**

- No null or missing values were detected in any column.

5. **Unique Value Analysis:**

- **Category:** **Technical, Billing, Feature Request, Account Management**
→ consistent.

- **Priority:** High, Urgent, Medium, Low → consistent.
- **Status:** Closed, Resolved → no other variations detected.
- **Tags:** login-issue, server-error, payment-failed, account-changes, refund, feature, design, incorrect-billing, connection-error → no missing or unexpected tags, though multi-label tags could be standardized.
- **Sentiment:** Observed multiple representations of neutral sentiment ('Neutral', 'Neutral ' with space) → inconsistency detected.

Summary of Detected Issues:

- Date columns stored as text rather than datetime.
- Minor inconsistency in sentiment labeling.
- No missing or duplicate rows, and core fields were largely consistent.

This evaluation provides an overview of **potential data issues that could impact analysis** while leaving the actual corrections for the next stage.

4. Data Cleaning Process

This section shows how the data cleaning process and its systematic steps were carried out to ensure that the Freshdesk Customer Support Tickets Dataset is well-structured, standardized, and ready for analysis. The following steps were performed to prepare the dataset for further insights:

1. Removal of Irrelevant or Sensitive Columns: Columns not necessary for analysis were reviewed, ensuring focus on core analytical fields such as Ticket ID, Category, Priority, Status, Tags, Sentiment, and timestamps. No columns were identified as necessary for removal.

Loading Dataset

```
[1] ✓ 21s
# Importing necessary libraries
import pandas as pd
import numpy as np

# Google Drive
from google.colab import drive
drive.mount('/content/drive')

# Defining the file path
file_path = '/content/drive/MyDrive/Customer Insights and Data Analytics Early Remote Internship/Freshdesk Customer Support Tickets Dataset Filled.csv'

# Reading the CSV file
df = pd.read_csv(file_path)

# Displaying first few rows to verify
df.head()
```

Mounted at /content/drive

	Ticket ID	Created Date	Resolved Date	Agent	Customer	Subject/Short Description	Category	Priority	Status	Tags	Sentiment
0	T011	2026-01-04 13:30	2026-01-04 15:00	Agent B	Customer K	Account locked after multiple attempts	Technical	High	Closed	login-issue	Negative
1	T012	2026-01-04 14:15	2026-01-04 16:00	Agent C	Customer L	Subscription auto-renewal failed	Billing	Urgent	Resolved	payment-failed	Neutral
2	T013	2026-01-05 09:00	2026-01-05 11:20	Agent A	Customer M	Error during checkout	Technical	Urgent	Resolved	server-error	Negative
3	T014	2026-01-05 09:45	2026-01-05 10:30	Agent B	Customer N	Request for multiple language support	Feature Request	Low	Closed	feature	Positive
4	T015	2026-01-05 11:00	2026-01-05 12:50	Agent C	Customer O	Cannot change profile picture	Account Management	Medium	Resolved	account-changes	Neutral

Next steps: [Generate code with df](#) [New interactive sheet](#)

Data Extraction & Initial Assessment

```
[21]
✓ 0s
# Checking number of rows and columns
rows, columns = df.shape
print(f"Number of rows: {rows}")
print(f"Number of columns: {columns}")
```

... Number of rows: 30
Number of columns: 11

2. Date & Time Formatting: Other column's datatypes were perfect but **Created Date** and **Resolved Date** were converted from text to **datetime objects** to ensure accurate time-based calculations and compatibility with visualization tools. *Before vs After (Screenshots)*

```
[22]
✓ 0s
# Checking data types of all columns
df.dtypes
```

Ticket ID	object
Created Date	object
Resolved Date	object
Agent	object
Customer	object
Subject/Short Description	object
Category	object
Priority	object
Status	object
Tags	object
Sentiment	object

dtype: object

```
[23]
✓ 0s
# Converting 'Created Date' and 'Resolved Date' to datetime
df['Created Date'] = pd.to_datetime(df['Created Date'], format='%Y-%m-%d %H:%M')
df['Resolved Date'] = pd.to_datetime(df['Resolved Date'], format='%Y-%m-%d %H:%M')

# Verify the conversion
df.dtypes
```

Ticket ID	object
Created Date	datetime64[ns]
Resolved Date	datetime64[ns]
Agent	object
Customer	object
Subject/Short Description	object
Category	object
Priority	object
Status	object
Tags	object
Sentiment	object

dtype: object

3. Handling of Missing Values & Duplicate Rows: Each column was checked for nulls. No missing values were detected in this dataset & no duplicate rows were found , ensuring completeness.

```
[3]
✓ 0s
# Checking for duplicate rows
duplicate_rows = df[df.duplicated()]
print(f"Number of duplicate rows: {duplicate_rows.shape[0]}")
```

... Number of duplicate rows: 0

```
[4]
✓ Os ▶ # Checking for null values in each column
null_counts = df.isnull().sum()
print("Null values in each column:")
print(null_counts)
```

```
✓ ... Null values in each column:
Ticket ID          0
Created Date       0
Resolved Date      0
Agent              0
Customer           0
Subject/Short Description 0
Category           0
Priority           0
Status            0
Tags              0
Sentiment          0
dtype: int64
```

4. Inspection of Key Categorical Columns: Unique values in **Category**, **Priority**, **Status**, **Tags**, and **Sentiment** were examined to detect inconsistencies, typos, or variations in naming conventions. This step helped identify issues such as extra spaces in **Sentiment** values, ensuring all categorical fields were standardized.

```
[24]
✓ Os ▶ # Check unique values in key categorical columns
categorical_columns = ['Category', 'Priority', 'Status', 'Tags', 'Sentiment']

for col in categorical_columns:
    print(f"Unique values in '{col}':")
    print(df[col].unique())
    print("-" * 50)
```

```
✓ ... Unique values in 'Category':
['Technical' 'Billing' 'Feature Request' 'Account Management']
-----
Unique values in 'Priority':
['High' 'Urgent' 'Low' 'Medium']
-----
Unique values in 'Status':
['Closed' 'Resolved']
-----
Unique values in 'Tags':
['login-issue' 'payment-failed' 'server-error' 'feature' 'account-changes'
 'refund' 'design' 'incorrect-billing' 'connection-error']
-----
Unique values in 'Sentiment':
['Negative' 'Neutral ' 'Positive' 'Neutral']
-----
```

```
[25]
✓ Os # Stripping whitespace and ensuring consistent capitalization in 'Sentiment'
df['Sentiment'] = df['Sentiment'].str.strip().str.capitalize()

# Checking again
print(df['Sentiment'].unique())
```

```
✓ ['Negative' 'Neutral' 'Positive']
```

5. Export of Cleaned Dataset: After all cleaning and standardization steps, the finalized dataset was exported as “**Cleaned Freshdesk Customer Support Tickets Dataset Filled**” in the same Google Drive folder to ensure it is ready for further analysis.

```
[26]: # Exporting csv file
      output_path = "/content/drive/MyDrive/Customer Insights and Data Analytics Early Remote Internship/Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled.csv"
      # Saving as CSV
      df.to_csv(output_path, index=False)
      print(f"Cleaned dataset successfully saved to: {output_path}")
```

... Cleaned dataset successfully saved to: /content/drive/MyDrive/Customer Insights and Data Analytics Early Remote Internship/Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled.csv

→ **Exported Cleaned Dataset (link):**

[Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled.csv](#)

→ To view the python code (jupyter notebook in google colab) which we applied for data cleaning, click here: [Ticket_Data_Cleaning.ipynb](#)

We systematically cleaned and standardized the dataset to ensure that all ticket information is accurate, consistent, and well-structured. By verifying categories, priorities, statuses, tags, sentiments, and date formats, we prepared the dataset for reliable analysis, visualization, and reporting. Our Python-based cleaning process enabled us to verify each step through immediate outputs, ensuring data integrity and consistency. This foundational work is essential for uncovering meaningful customer support insights and supporting data-driven decision-making in subsequent sections.

5. Tagging & Categorization Strategy

This section details the tagging & categorization strategy of Freshdesk customer support ticket dataset. Categories help us identify which team we must contact to help resolve the issue. One department can handle a wide range of different issues, and categories alone cannot help us precisely identify the top-most recurring issues. To remedy this, we use tags, think of them as sub-categories, to help us gain better insight and detail.

5.1 How Tagging Improves Categorization and Visibility

Tags help highlight what issues are **truly** prevalent. We could have the most issues come from the technical category, but the most recurring tag is payment-issues. Tags would help highlight that our most common issue is payment; without it, we would've focused on technical instead.

Tagging also adds depth to our categories. If a lot of our issues come from billing, that information is not enough to help us know what needs to be improved. Do we need to implement a smoother refund process? Improve our payment system? Pay more attention to

invoice details? Without tags, we would be unclear on what specific improvements need to be made.

5.2 Category-wise Tag Definitions

To ensure consistent and meaningful tagging, each category was assigned a predefined set of tags that represent the most common and relevant sub-issues within that category. These tags were designed to capture recurring problem types and provide deeper visibility into customer concerns. The following points outline the primary tags used for each category and their respective purposes.

★ **Technical category tags**

- *Login-issue*: this includes issues related to account access, ranging from accounts being locked to a customer forgetting their password.
- *Server-error*: issues that point to bugs and errors in the system.
- *Connection-error*: occurs due to a bad internet connection. It could result from the customer having connection issues or the servers being down.

★ **Billing category tags**

- *Refund*: issues where a customer requires a refund
- *Payment-failed*: issues where a payment has failed to go through
- *Incorrect-billing*: issues where a customer has been charged the incorrect amount

★ **Feature Request category tags**

- *Feature*: these are requests for features that a customer can interact with
- *Design*: these are requests for visual changes to make the app better on the eyes.

★ **Account Management category tags**

- *Account-changes*: this is where a customer wants to edit details about their account but may be unable to navigate the app.

Each category has its own set of tags; however, sometimes tags used for one category may be applied to a different category. This is because even though the tag is appropriate, the category the tag originates from is not the correct team to contact. An example of this is ticket T017, the user wants to reset their password to log in, making this a login issue; however, this issue does not need to be passed to the technical team. The account management team can explain how to receive a new link, as they typically handle navigation issues.

6. Manual Sentiment Classification Approach & Evaluation

This section explains how customer sentiment was identified in the dataset. The goal of this step is to apply sentiment labels in a clear and consistent way so the data can be used confidently in later analysis.

Sentiment was classified manually using simple rules, not automated tools or software. Each ticket was reviewed one at a time. The “Subject / Short Description” column was the main field used to understand how the customer felt, while the “Tags” column was used to support the decision when more context was needed.

The following simple rules were used to classify sentiment:

Sentiment Class	Typical Subject / Issue Type	Indicative Tags	Classification Method & Rationale
Positive	Feature requests, design improvements, usability suggestions	feature, design	We classified a ticket as positive when the overall tone sounded constructive and forward-looking. The customer usually proposed ideas, enhancements, or improvements rather than reporting problems. The language appeared polite, calm, and collaborative, often focusing on how the product could be better instead of expressing dissatisfaction. There was no indication of urgency, frustration, or service failure, which signaled a generally positive attitude toward the platform.
Neutral	General inquiries, routine account changes, access-related requests without complaints	account-changes, login-issue (non-error cases)	We identified neutral sentiment when the message sounded factual and informational. The customer described a requirement, question, or request in a straightforward manner without emotional wording. The tone neither praised nor criticized the service and did not indicate inconvenience or satisfaction.

			Such tickets were treated as neutral because they reflected normal usage communication rather than an emotional response.
Negative	Technical issues, system errors, billing disputes, payment failures, access problems	login-issue, server-error, payment-failed, incorrect-billing, refund, connection-error	We classified a ticket as negative when the wording conveyed frustration, inconvenience, or disruption. Messages reporting errors, failures, blocked access, or financial issues often carried an urgent or complaining tone. The language suggested dissatisfaction with service performance or outcomes, indicating that the customer experience was negatively affected at the time of submission.

Using this rule-based manual approach helped ensure that sentiment labels were applied consistently across all tickets, based on the overall message and intent of the customer.

7. Sentiment Analysis Summary

This section summarizes the sentiment analysis. After classifying sentiment, clear patterns were observed across different types of issues. Tickets with;

- **Negative sentiments** were mainly linked to login problems, system errors, billing issues, and refund requests, where customers were experiencing service difficulties.
- **Neutral sentiment** appeared most often in routine account management requests and informational questions that did not involve major problems.
- **Positive sentiment** was mostly found in feature requests and design or user interface suggestions, where customers focused on improving the product rather than reporting issues.

These patterns show that customer sentiment closely reflects the type of issue being reported and adds useful context to category and tag-based analysis.

8. Key Observations & Insights

This section shows the key observations and insights we derived from the data cleansing, tagging, categorization, and sentiment analysis processes.

8.1 Challenges Faced during Data Cleansing

- The columns for Created Date and Resolved Date were in text form and not in datetime format.
- Some minute discrepancies were noticed in the categorical variables, particularly the extra spaces in the Sentiment column.
- The first importation of data into all columns was carried out as object variables.
- No missing or repeated values were observed in the dataset.

8.2 Insights Gained from Tagging and Categorization Improvement

- The technical problems were predominantly reported, and most were related to high and urgent priority tickets.
- Login-issue problems also appeared in the Technical and Account Management categories.
- Tagging allowed for more specific identification of issues like 'payment failed' or 'server error'.
- The feature request tickets were dominantly linked with low priority levels.

8.3 Sentiment Identification using Sentiment Analysis

- Mostly, negative feelings were identified to link with billing problems, failed payments, refund, and technical issues.
- The account maintenance and information requests were done with a neutral tone.
- The positive suggestions were primarily related to request features and design suggestions.
- Customer sentiment was very much in line with the type of problem identified.

9. Learning Outcomes

- This week helped us understand better how careful data extraction lays the groundwork for meaningful and trustworthy analysis.
- Through hands-on cleaning, we realized how small improvements such as standardizing dates, priorities, and categories, can enhance data quality.
- Working with structured tags made us appreciate how clear labeling reduces confusion and makes customer issues easier to interpret.
- Applying manual, rule-based sentiment classification sharpened my awareness of how tone and wording in tickets reflect customer experience.
- Overall, this week strengthened our confidence in preparing raw support data into a well-organized and analysis-ready form.

10. Conclusion

In Week-2, we successfully focused on preparing a clean, structured, and reliable customer support ticket dataset through systematic data extraction, cleaning, tagging, and manual sentiment classification. The standardized priorities, categories, tags, and sentiment labels significantly improved data consistency and interpretability. This enhanced data quality ensures more accurate analysis and meaningful insights in subsequent stages. Overall, the dataset is now fully analysis-ready and well-suited for Week-3 activities, including trend identification, SLA evaluation, volume analysis, and dashboard visualization.

End of Report

Thank You !